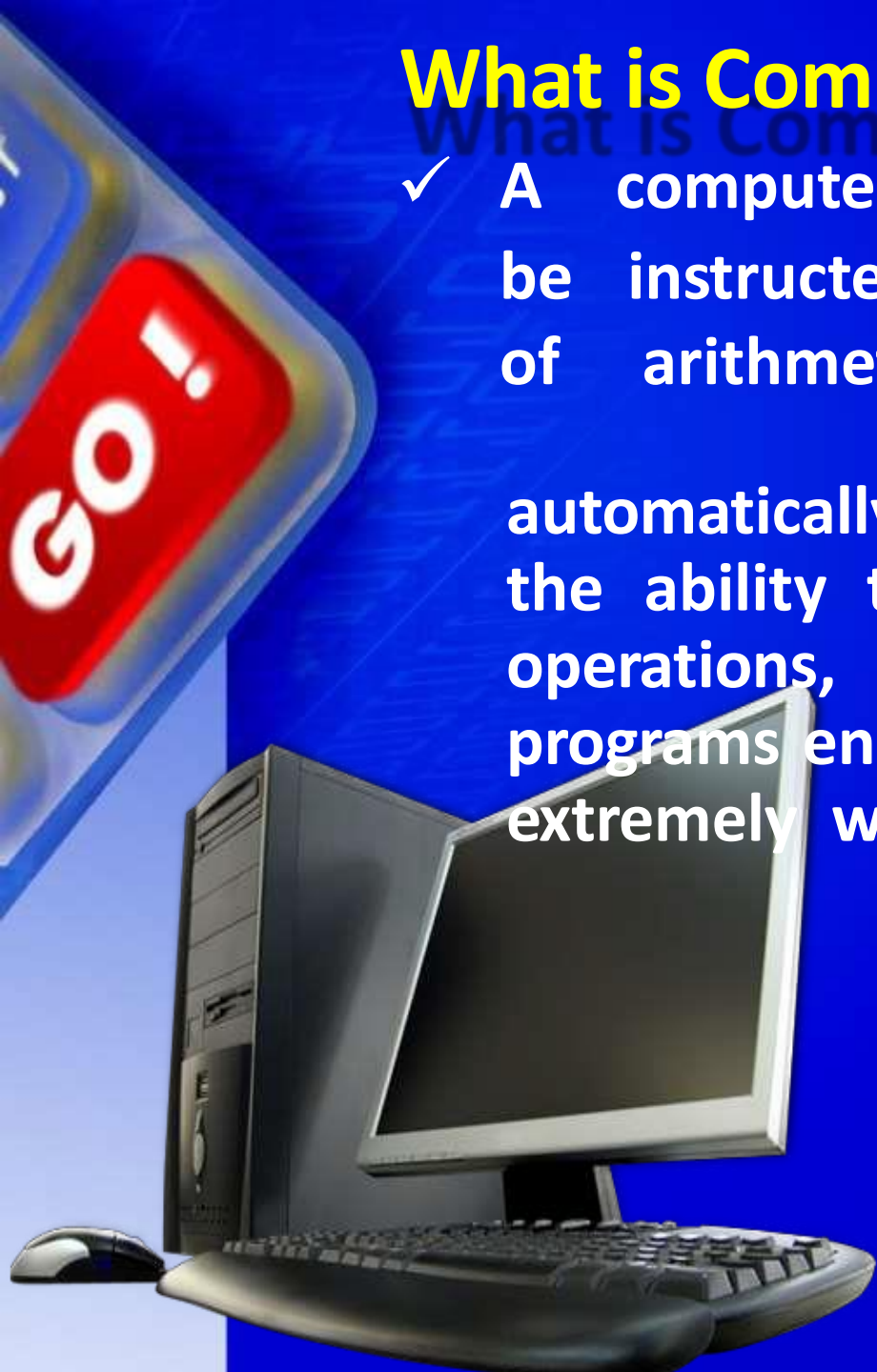




COMPUTER FUNDAMENTALS I

What is Computer?

- ✓ A computer is a device that can be instructed to carry out sequences of arithmetic or logical operations automatically. Modern computers have the ability to follow generalized sets of operations, called *programs*. These programs enable computers to perform an extremely wide range of tasks.





What is Data?

- ✓ Data. Data in a computer terminology mean raw facts and figures.



What is Information?

What is Information?



- ✓ means what we get after processing.



Data Vs. Information?

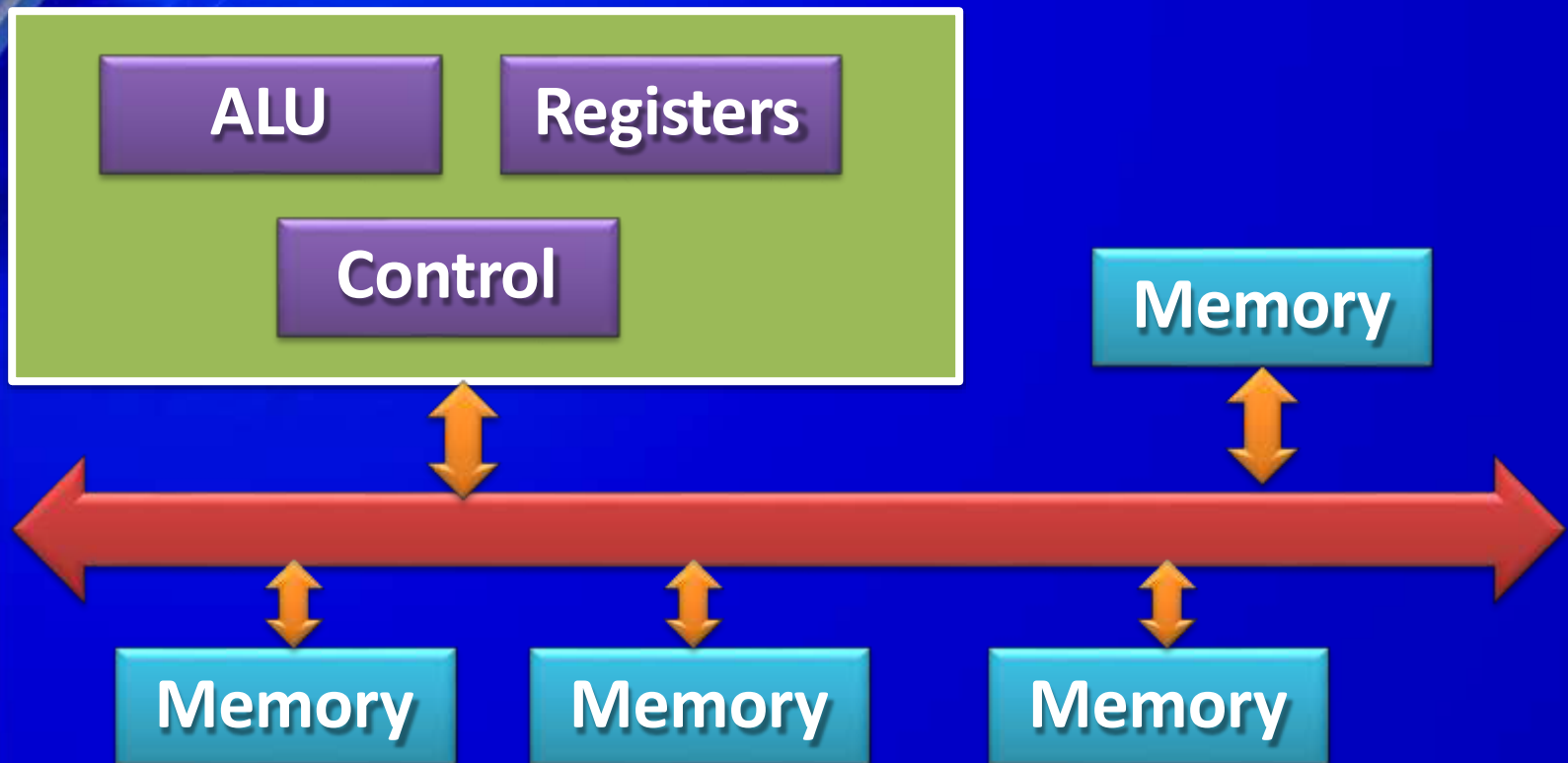
- ✓ you should understand that what goes in the computers is data and what comes out of them is information.
- ✓ This process of turning data into information is also known as information processing cycle.



BASIC COMPUTER ORGANISATION

BASIC COMPUTER ORGANISATION

Computer organisation refers to logical structure of a computer how its components are connected to one another



Functional components of a Computer

✓ Input Unit

✓ CPU

✓ Output Unit

✓ Memory



Input Unit

- ✓ The input unit consists of input devices that are attached to the computer.
- ✓ These devices take input and convert it into binary language that the computer understands.
- ✓ Some of the common input devices are keyboard, mouse, joystick, scanner etc.



Input Unit

Input Devices of Computer



Touch screen

Camera



Scanner



Microphone



Joystick



Mouse

www.examplesof.net



Keyboard



Web cam



Track ball



Central Processing Unit(CPU)

- ✓ Once the information is entered into the computer by the input device, the processor processes it.
- ✓ The CPU is called the brain of the computer because it is the control centre of the computer.
- ✓ As the CPU is located on a small chip, it is also called the Microprocessor.



Central Processing Unit(CPU)

- The CPU has three main components which are responsible for different functions –
 - Arithmetic Logic Unit (ALU),
 - Control Unit (CU) &
 - Memory registers.



Arithmetic Logic Unit (ALU)

- ✓ The ALU, as its name suggests performs mathematical calculations and takes logical decisions. Arithmetic calculations include addition, subtraction, multiplication and division.
- ✓ Logical decisions involve comparison of two data items to see which one is larger or smaller or equal.



CU Control Unit (CU)

- ✓ The Control unit coordinates and controls the data flow in and out of CPU and also controls all the operations of ALU, memory registers and also input/output units.
- ✓ It is also responsible for carrying out all the instructions stored in the program.

MR Memory Registers

- ✓ A register is a temporary unit of memory in the CPU.
- ✓ These receive data/information and then this data/information is held in them as per the requirement.

Output Unit

- ✓ The output unit consists of output devices that are attached with the computer.
- ✓ It converts the binary data coming from CPU to human understandable form. The
- ✓ common output devices are monitor, printer, plotter etc.



Classification of Computers



Classification of Computers

✓ The computers can be classified based on the technology being used as:

✓ Digital Computers

✓ Analog Computers &

✓ Hybrid Computers

Digital Computers

- 
- ✓ These computers are capable of processing information in discrete form.
 - ✓ In digital technology data which can be in the form of letters, symbols or numbers is represented in binary form i.e. 0s and 1s.

Digital Computers

- ✓ Binary digits are easily expressed in a digital computer by the presence (1) or absence (0) of current or voltage.
- ✓ It computes by counting and adding operations.

Digital Computers

- 
- ✓ The digital computers are used in industrial, business and scientific applications.
 - ✓ They are quite suitable for large volume data processing.

Digital Computers

EXAMPLES:

- ✓ IBM PC, HP, LENOVO, DELL, ...etc
- ✓ Apple Macintosh
- ✓ Calculators
- ✓ Digital watches etc

Analog Computers

- 
- ✓ An Analog computer works on continuously changeable aspects of physical phenomenon such as fluid pressure, mechanical motion and electrical quantities.
 - ✓ These computers measure changes in continuous physical quantities say current and voltage.

Analog Computers

Examples

- ✓ Thermometer.
- ✓ Analog clock.
- ✓ Speedometer.
- ✓ Tire pressure gauge.

Hybrid Computers

- ✓ These use both analog and digital technology.
- ✓ It has the speed of analog computer and the accuracy of a digital computer.
- ✓ It may accept digital or analog signals but an extensive conversion of data from digital to analog and analog to digital has to be done.

Hybrid Computers



✓ Hybrid Computers are used as a cost effective means for complex simulations.

✓ **Examples:** Computer used in hospitals to measure the heartbeat of the patient. Devices used in petrol pump.

Hybrid Computers

EXAMPLES OF HYBRID COMPUTERS



Classification of Digital Computers

- 
- ✓ The digital computers are classified according to their computing capabilities.



Classification of Digital Computers

✓ The various types of digital computers are :

✓ Micro Computers

✓ Main Frames

✓ Super Computers

Micro Computers

- ✓ These are also known as Personal Computers. These type of digital computer uses a microprocessor (a CPU on a single chip) and include both desktops and laptops.

Micro Computers



✓ These computers can work on small volume of data, are very versatile and can handle variety of applications. These computers are being used as work stations, CAD, multimedia and advertising applications.

Micro Computers



Micro Computers



Mainframe Computers

- ✓ These computers are large and very powerful computers with very high memory capacity.
- ✓ These can process huge databases such as census at extremely fast rate. They are suitable for big organizations, banks, industries etc. and can support hundreds of users simultaneously on the network.

Super Computers

- ✓ These are fastest and very expensive computers.
- ✓ They can execute billions of instructions per second.

Super Computers



✓ These are multiprocessor, parallel systems suitable for specialized complex scientific applications involving huge amounts of mathematical applications such as weather forecasting.

Super Computers

- 
- ✓ The main difference between a supercomputer and a mainframe is that a supercomputer executes fewer programs as fast as possible whereas a mainframe executes many programs concurrently.



Hardware and Software



Hardware and Software

A computer consists of both hardware and software and both are equally important for the working of the computer system.

✓ The electronic components of a computer system that we can see and touch are called **hardware**.

✓ **Software** is a general term used for computer programs that control the operations of the computer.



Hardware and Software

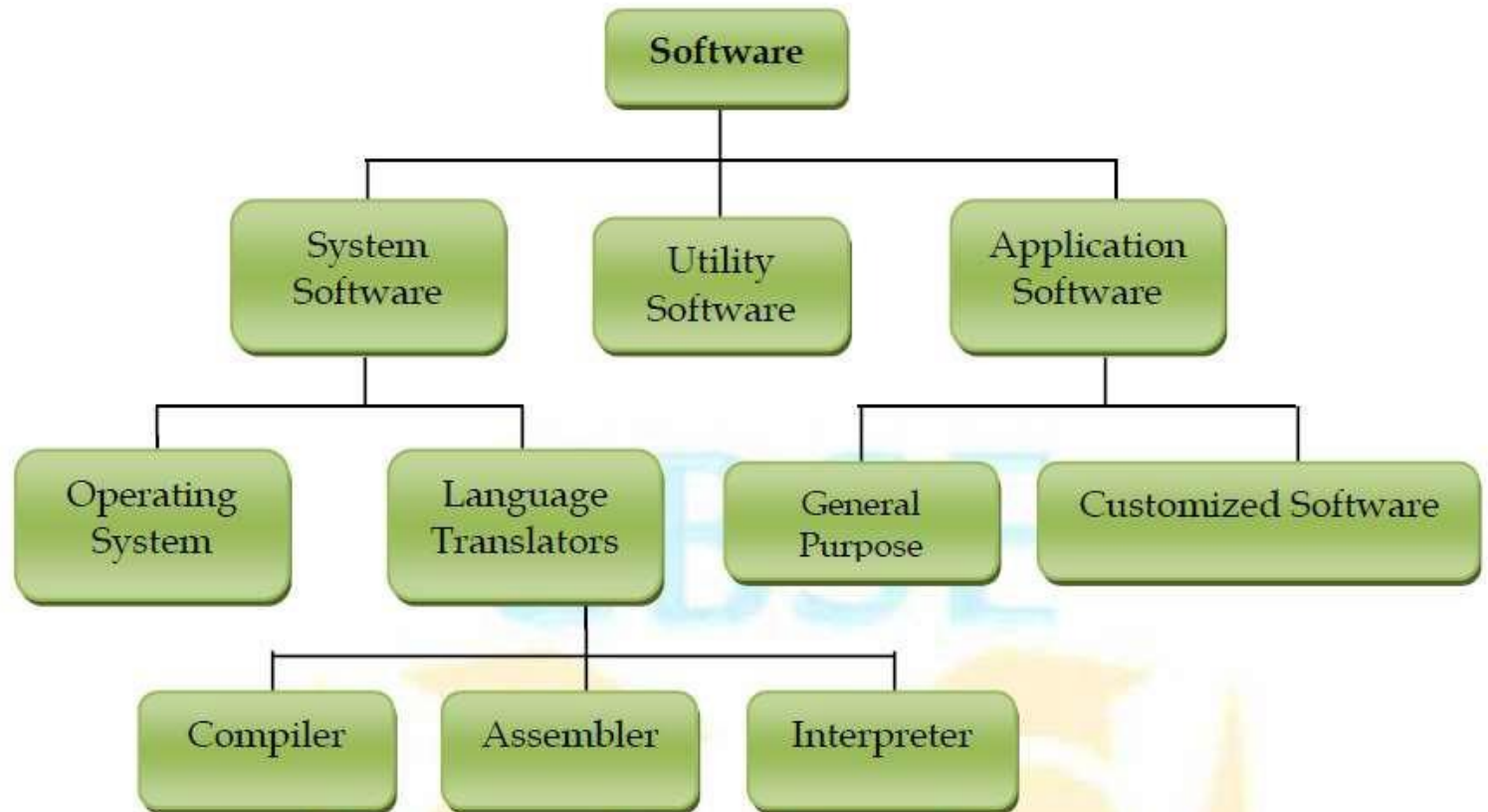
- ✓ A program is a sequence of instructions that perform a particular task. A set of programs form a software.
- ✓ It is the software which gives hardware its capability.
- ✓ Hardware is of no use without software and software cannot be used without hardware.

Types of Software

Software can be broadly are categorized as:

- ✓ **System Software**
- ✓ **Application Software**
- ✓ **Utility Software**

Types of Software





System Software

System Software is the software that is directly related to coordinating computer operations and performs tasks associated with controlling and utilizing computer hardware. These programs assist in running application programs and are designed to control the operation of a computer system.



System Software

System software directs the computer what to do, when to do and how to do. System software can be further categorized into:-

- ✓ **Operating System**
- ✓ **Language Translators**



Operating System

- ✓ An Operating system is the most important system software.
- ✓ It is a set of programs that control and supervise the hardware of a computer and also provide services to application software, programmers and users.

Operating System

- 
- ✓ Some of the popular operating systems used in personal computers are DOS, Windows, Unix, Linux, Solaris, Android , etc.



Application Software

- ✓ An application software is bought by the user to perform specific applications or tasks.
- ✓ Say for example making a document or making a presentation or handling inventory or managing the employee database.



Application Software

✓ An application software can be of two types:

✓ General Purpose Application Software &

✓ Customized Application software.



Application Software - General Purpose Application Software

- ✓ Some of the application software is made for the common users for day to day applications and uses.
- ✓ These are also referred as Office Tools.
- ✓ The users may use them in the manner they want.



Application Software - General Purpose

Application Software

- ✓ **Some of the popular types of general purpose application software are:**
- ✓ **Word Processor**
- ✓ **Presentation Tools**
- ✓ **Spreadsheet Packages**
- ✓ **Database Management System**



Application Software - Customized Software

- ✓ Customized Software is one which is tailor made as per the user's requirement. Such type of software is customer specific.



Application Software - Customized Software

Examples:

- ✓ Inventory Management System & Purchasing System
- ✓ School Management System
- ✓ Payroll System
- ✓ Financial Accounting
- ✓ Hotel Management
- ✓ Reservation System
- ✓ Weather Forecasting system



MEMORY CONCEPTS



Memory

- ✓ Memory is one of the most important components of a computer system as it stores data and instructions.
- ✓ Every memory chip contains thousands of memory locations. In the computer, the data is stored in the form of bits and bytes.

Memory

- ✓ A bit is the smallest storage unit of memory. A nibble is a collection of 4 bits. Eight bits combined together to form a single byte, which in turn represents a single character

Memory Units

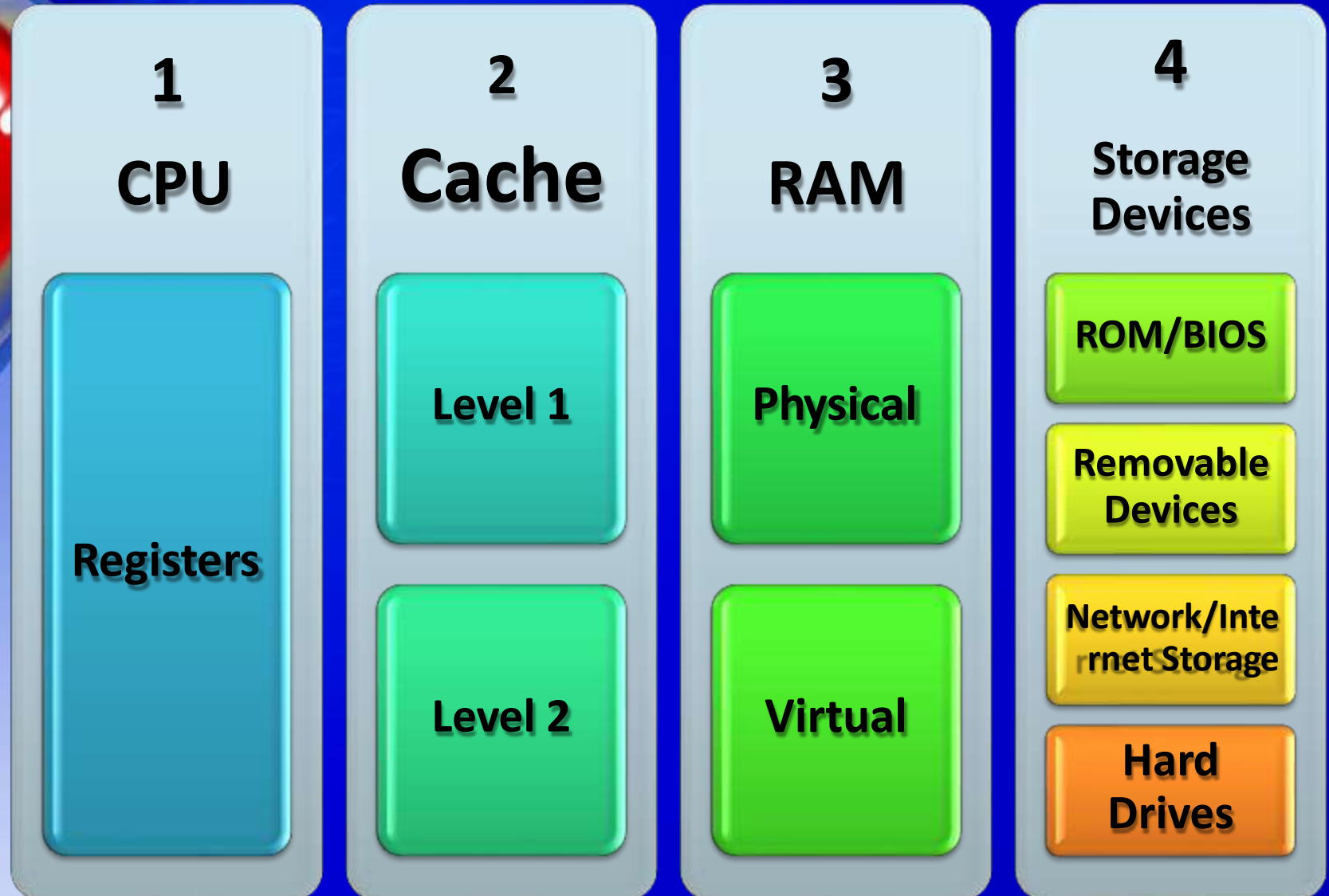
Memory unit	Relationship with earlier memory unit	In equivalent Bytes
Kilo Byte (KB)	1 Kilo Byte = 1024 Bytes(or 2^{10} Bytes)	1024
Mega Byte (MB)	1 Mega Byte = 1024 Kilo Byte(or 2^{10} KB)	1024x1024
Giga Byte (GB)	1 Giga Byte = 1024 Mega Byte(or 2^{10} MB)	1024x1024x1024
Tera Byte (TB)	1 Tera Byte = 1024 Giga Byte(or 2^{10} GB)	1024x1024x1024x1024
Peta Byte (PB)	1 Peta Byte = 1024 Tera Byte(or 2^{10} TB)	1024x1024x1024x1024x 1024
Exa Byte(EB)	1 Exa Byte = 1024 Peta Byte(or 2^{10} PB)	1024x1024x1024x1024x 1024x1024

Memory Units

Zetta Byte(ZB)	1 Zetta Byte = 1024 Exa Byte(or 2^{10} EB)	1024x1024x1024x1024x 1024x1024x 1024
Yotta Byte(YB)	1 Yotta Byte = 1024 Zetta Byte(or 2^{10} ZB)	1024x1024x1024x1024x 1024x1024x 1024x1024

Temporary Storage areas

Permanent Storage areas



Types of Memories

The computer memories can be divided into following categories:

- ✓ **Primary Memory**
- ✓ **Secondary memory**
- ✓ **Cache Memory**



Primaryy Memory



Primary Memory

Primary memory or main memory is a Metal Oxide Semiconductor (MOS) memory used for storing program and data during the execution of the program. It is directly accessible to CPU.



Primary Memory

Broadly primary memory can be of two types :—

1. RAM (Random Access Memory)

2. ROM (Read only memory).





Secondary Memory



Secondary Memory

- ✓ The major limitation of primary memory is that it has limited storage capacity and is volatile.
- ✓ To overcome this limitation we have secondary memory storage devices. This type of memory is also called external memory. For example Floppy disk, hard disk, USB drives, CD/DVDs



Hard Disk Storage





Thank You